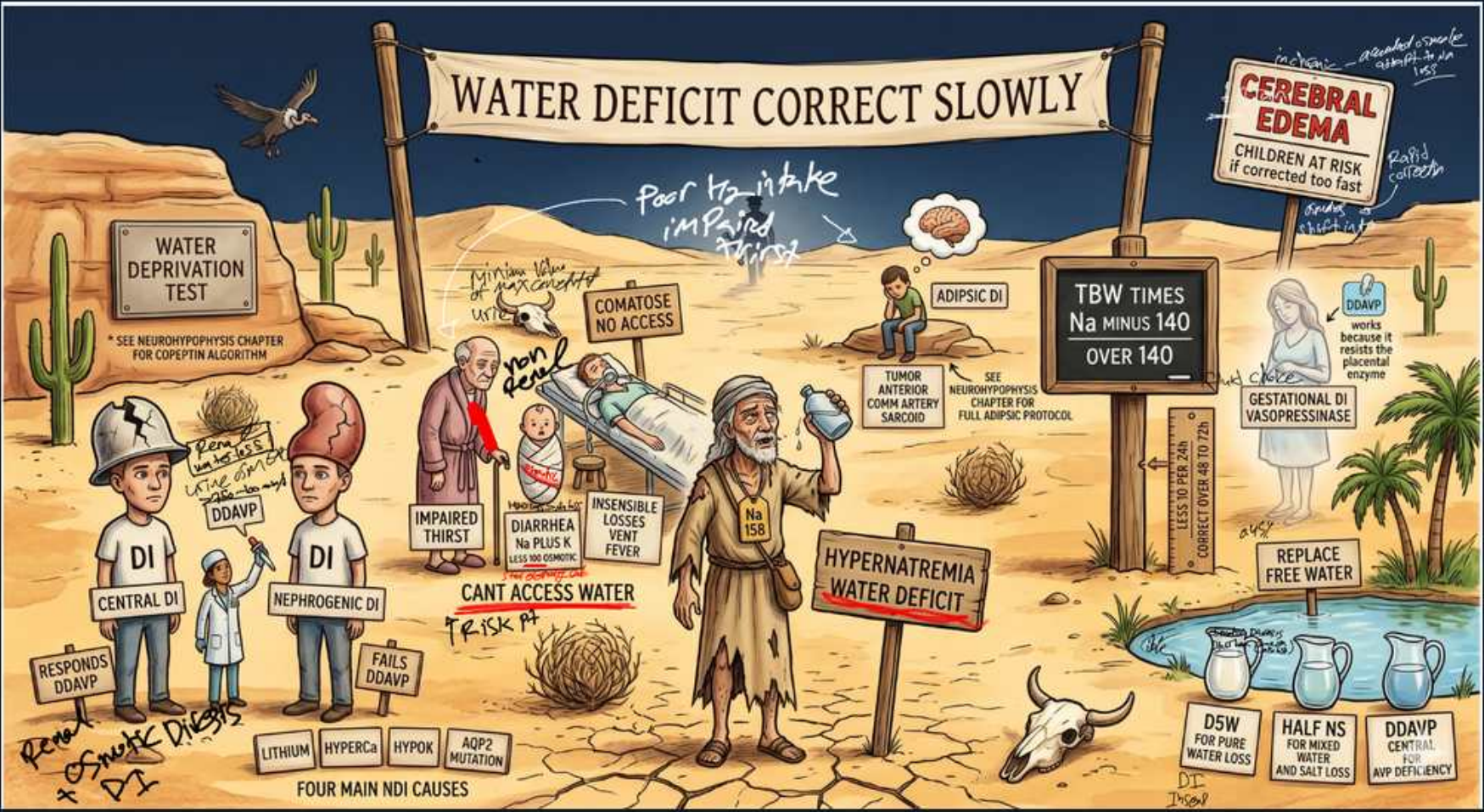


Hypernatremia — Causes & DI Overview



1. Water deficit

WHAT YOU SEE

Banner: WATER DEFICIT CORRECT SLOWLY over a desert

WHAT IT ENCODES

Hypernatremia is DEFINED as serum Na >145 mEq/L and is ALWAYS a hypertonic, hyperosmolar state (serum osm >295 mOsm/kg) — there is no hypotonic or isotonic hypernatremia, unlike hyponatremia. The core problem is a deficit of total body water RELATIVE to sodium: TBW falls, so the same Na is dissolved in less water and the concentration climbs. Estimate the gap with the free water deficit = TBW × (Na/140 – 1), where TBW ≈ 0.6 × weight (men) or 0.5 × weight (women). The banner says 'correct slowly' because the brain adapts to the hypertonic state.

BOARD TRAP

WRONG answer: 'hypernatremia is caused by eating too much salt.' Pure sodium GAIN is the rarest mechanism (hypertonic saline, NaHCO3 during a code, hypertonic tube feeds) — the vast majority of cases are WATER LOSS or lack of water intake. Discriminator: a true salt-gain patient is hypervolemic/edematous with high urine Na, whereas a water-loss patient is hypovolemic with dry mucous membranes.

2. Impaired thirst/access

WHAT YOU SEE

Old man in robe holding water he cannot drink; 'CANT ACCESS WATER'

WHAT IT ENCODES

Sustained hypernatremia requires a SECOND hit beyond water loss: an impaired thirst mechanism OR no access to water. A normal hypothalamus senses a rise of just ~1–2% in osmolality and triggers intense thirst, so an awake person with a glass of water self-corrects within hours. Therefore persistent hypernatremia is essentially a disease of the very young (infants who cannot ask for water), the very old (blunted thirst), and the dependent (intubated, sedated, restrained, demented, post-stroke). The old man holding water he cannot drink is the visual mnemonic for 'the water is there but the patient can't get it in.'

BOARD TRAP

WRONG answer: 'a healthy, alert outpatient with hypernatremia just needs IV fluids and no further workup.' If an awake patient with free access to water stays hypernatremic, something is wrong with thirst sensing (adipsic/hypodipsic hypothalamic lesion) — investigate it. Discriminator: ask whether the patient is physically and cognitively able to drink; if yes, suspect a central thirst defect.

3. Comatose/no access

WHAT YOU SEE

Comatose patient in bed, 'COMATOSE NO ACCESS'

WHAT IT ENCODES

The comatose patient in bed is the prototype of HOSPITAL-ACQUIRED hypernatremia: intubated, sedated, post-op, comatose, or dependent patients physically cannot drink and rely entirely on what the team prescribes. Insensible losses (fever, ventilator, tachypnea) plus inadequate free-water orders push Na up over days. This is the most common iatrogenic cause on inpatient wards and is largely preventable by ordering maintenance free water (oral/NG water or D5W).

BOARD TRAP

WRONG answer: 'the ICU patient's rising Na must mean they are getting too much sodium.' In ventilated/comatose patients the usual driver is inadequate free-water REPLACEMENT against ongoing insensible and renal losses, not sodium overload. Discriminator: review the I/O and free-water balance — the deficit is on the water side.

4. Impaired thirst

WHAT YOU SEE

Person labeled IMPAIRED THIRST

WHAT IT ENCODES

Hypodipsia/adipsia results from damage to the hypothalamic osmoreceptors that sense tonicity and drive thirst. Because the brake on water deficit is gone, even modest losses go uncorrected and Na drifts up. Causes mirror central DI lesions — hypothalamic tumors (craniopharyngioma, germinoma), trauma, surgery, sarcoidosis/granulomatous disease, and anterior communicating artery aneurysm/rupture. Some patients have combined adipsia + central DI ('adipsic DI'), the most dangerous form.

BOARD TRAP

WRONG answer: 'just tell the patient to drink more.' Patients with true adipsia do not FEEL thirsty, so they will not drink to thirst — they need a scheduled fixed daily water prescription and weight-based monitoring. Discriminator: a patient who denies thirst despite a measured high serum osmolality has an osmoreceptor/thirst defect, not simple dehydration.

5. Adipsic DI

WHAT YOU SEE

Thought bubble 'ADIPSIC DI' with tumor note

WHAT IT ENCODES

Adipsic central DI is the worst-case combination: a hypothalamic lesion destroys BOTH AVP-producing neurons (no urine concentration → water loss) AND the thirst center (no compensatory drinking). The result is wildly swinging, often severe hypernatremia. Classic causes are hypothalamic tumors, anterior communicating artery aneurysm clipping/rupture, neurosarcoidosis, and trauma. These patients require a fixed daily fluid prescription plus desmopressin, with frequent Na and daily weight checks because they cannot self-regulate.

BOARD TRAP

WRONG answer: 'manage adipsic DI by titrating DDAVP to symptoms and letting thirst guide fluid intake.' Thirst is absent, so you cannot rely on it — these patients overshoot or undershoot dangerously and need protocolized weight-based water dosing. Discriminator: adipsic DI = no thirst + dilute urine + hypothalamic lesion on imaging.

6. Insensible/GI loss

WHAT YOU SEE

'INSENSIBLE LOSSES', 'DIARRHEA Na PLUS K' note

WHAT IT ENCODES

These are HYPOTONIC fluid losses — you lose water in excess of sodium, so the remaining serum becomes more concentrated. Extrarenal routes: GI (osmotic diarrhea, vomiting, NG suction, lactulose), skin/sweat (fever, burns, heat), and insensible/respiratory (tachypnea, ventilator, fever — roughly 10 mL/kg/day baseline, more with fever). In all of these the kidney is working correctly and concentrates urine maximally (Uosm >700–800), so the diagnostic clue is high urine osmolality with low urine Na.

BOARD TRAP

WRONG answer: 'all diarrhea causes hyponatremia.' OSMOTIC diarrhea (lactulose, malabsorption, viral gastroenteritis) loses fluid that is hypotonic relative to plasma → hypernatremia, whereas SECRETORY diarrhea (cholera) is roughly isotonic and tends not to raise Na as much. Discriminator: check the type of diarrhea and the urine osmolality — extrarenal loss gives a maximally concentrated urine.

7. Central DI

WHAT YOU SEE

Child with 'DI' sign, CENTRAL DI, RESPONDS DDAVP

WHAT IT ENCODES

Central DI = deficient AVP (ADH) RELEASE from the posterior pituitary, so the collecting duct never gets the signal to insert aquaporin-2 channels → pure free-water loss with inappropriately dilute urine (Uosm often <300, can be <100). The defining test result: after giving desmopressin, urine osmolality RISES by >50% (typically ≥100% if complete) because the kidney is intact and just needed the hormone. Causes: idiopathic/autoimmune, pituitary surgery/trauma, tumors, infiltrative disease (sarcoid, Langerhans cell histiocytosis), and Sheehan/pituitary apoplexy.

BOARD TRAP

WRONG answer: 'a DI patient who concentrates urine after DDAVP has nephrogenic DI.' It is the opposite — a robust rise in Uosm after desmopressin defines CENTRAL DI (the kidney can respond once given AVP). Discriminator: central DI = LOW endogenous ADH but RESPONDS to exogenous DDAVP; copeptin (AVP surrogate) is low.

8. Nephrogenic DI

WHAT YOU SEE

Child with 'DI', NEPHROGENIC DI, FAILS DDAVP

WHAT IT ENCODES

Nephrogenic DI = the kidney is RESISTANT to AVP despite normal or high circulating levels, so the collecting duct fails to concentrate urine. Hallmark: little or no rise in urine osmolality (<50%, usually negligible) after desmopressin — the receptor/channel is the broken link, so adding more hormone does nothing. The lithium-treated patient who develops polyuria is the board's favorite vignette.

BOARD TRAP

WRONG answer: 'treat this AVP-resistant DI with high-dose DDAVP.' Desmopressin is largely INEFFECTIVE in nephrogenic DI because the defect is downstream of the hormone; you instead remove the offending drug and use a thiazide ± amiloride ± low-salt diet. Discriminator: nephrogenic DI = normal/high ADH (high copeptin) but FAILS to concentrate after DDAVP.

9. NDI causes

WHAT YOU SEE

Signs: LITHIUM, HYPERCA, HYPOK, AQP2 mutation

WHAT IT ENCODES

Acquired nephrogenic DI causes (the 'wanted poster' four): LITHIUM (most common — accumulates in principal cells via ENaC and downregulates AQP2), HYPERCALCEMIA (Ca sensing receptor activation impairs concentrating ability), and HYPOKALEMIA (impairs AQP2 expression). Hereditary causes are X-linked V2-receptor (AVPR2) mutations (~90% of congenital, boys) and autosomal AQP2 mutations. Also: chronic tubulointerstitial disease, obstruction relief, and demeclocycline.

BOARD TRAP

WRONG answer: 'lithium-induced nephrogenic DI always reverses immediately once the drug is stopped.' Long-standing lithium use can cause permanent tubulointerstitial damage and irreversible NDI even after discontinuation. Discriminator: correct hypercalcemia/hypokalemia (often reversible) but counsel that chronic lithium NDI may be only partially reversible — amiloride helps limit further uptake.

10. Water deprivation test

WHAT YOU SEE

Sign WATER DEPRIVATION TEST

WHAT IT ENCODES

The water deprivation test distinguishes the polyuria differentials: withhold water under supervision and watch urine osmolality. A normal person or primary polydipsic concentrates urine (Uosm rises appropriately, usually >600–800). DI patients FAIL to concentrate despite rising serum osm; you then give desmopressin — central DI now concentrates (>50% rise), nephrogenic DI does not. Copeptin assays are increasingly replacing the formal test.

BOARD TRAP

WRONG answer: 'perform a water deprivation test on a patient who is already hypernatremic to confirm DI.' If serum Na/osm is already high with dilute urine, the kidney has already 'failed' the deprivation — further withholding water is dangerous and unnecessary; just give DDAVP and watch the response. Discriminator: water deprivation is for the EUVOLEMIC, normonatremic polyuric patient — never deprive someone already hypertonic.

11. TBW × (Na–140)/140

WHAT YOU SEE

'TBW TIMES Na minus 140 OVER 140'

WHAT IT ENCODES

Free water deficit (liters) = $TBW \times (\text{serum Na}/140 - 1)$, where TBW ≈ 0.6 × weight in kg for men and 0.5 × for women (use 0.5 and 0.45 in the elderly). Example: a 70 kg man with Na 160 → $TBW 42 \text{ L} \times (160/140 - 1) = 42 \times 0.143 \approx 6 \text{ L}$ deficit. Replace this deficit PLUS ongoing obligatory and insensible losses, and remember the goal is to drop Na no faster than ~10 mEq/L per 24 h.

BOARD TRAP

WRONG answer: 'give the entire calculated free water deficit over the first few hours.' The deficit must be replaced gradually over ~48 hours (lowering Na ≤10 mEq/L/day) to avoid cerebral edema; the formula also ignores ongoing losses, which must be added. Discriminator: the calculation sets the TOTAL water to give, not the RATE — pace it to the safe correction limit.

12. Sodium gain

WHAT YOU SEE

'Na GAIN' / cattle skull near salt

WHAT IT ENCODES

Sodium gain is the LEAST common cause and produces hypervolemic hypernatremia WITHOUT a water deficit. Sources are almost always iatrogenic: hypertonic (3%) saline, repeated NaHCO₃ amps during a code, hypertonic tube feeds, and rarely massive salt ingestion or sea-water near-drowning. These patients are volume-EXPANDED with high urine Na, distinguishing them from the dehydrated water-loser.

BOARD TRAP

WRONG answer: 'treat sodium-gain hypernatremia with more isotonic saline.' Adding saline worsens the sodium load — the treatment is to remove the source and give free water (D5W), and if volume-overloaded/renal failure, a loop diuretic ± dialysis. Discriminator: hypervolemic + edematous + high urine Na = sodium gain, not water loss.

13. Replace free water

WHAT YOU SEE

Sign REPLACE FREE WATER near pool, D5W / half NS

WHAT IT ENCODES

Treatment is to replace FREE water: the safest route is enteral (oral or NG plain water); IV options are D5W (pure free water once dextrose is metabolized) and hypotonic saline ($\frac{1}{2}$ NS = 0.45% provides $\sim\frac{1}{2}$ free water). Choose based on hemodynamics — if the patient is in shock/hypovolemic, resuscitate with isotonic NS FIRST, then switch to free water once perfusion is restored. Calculate the rate to lower Na ≤ 10 mEq/L/24 h.

BOARD TRAP

WRONG answer: 'a hypotensive, severely volume-depleted hypernatremic patient should get D5W up front to bring the Na down fast.' Hemodynamic instability is corrected first with isotonic saline (NS) to restore perfusion; free-water replacement follows. Discriminator: shock $\rightarrow$ NS first; once stable $\rightarrow$ free water/D5W at the controlled correction rate.

14. Cerebral edema risk

WHAT YOU SEE

Sign CEREBRAL EDEMA — CHILDREN AT RISK if corrected too fast

WHAT IT ENCODES

In chronic (>48 h) hypernatremia the brain generates idiogenic osmolytes to pull water back in and restore its volume. If you then drop serum Na too fast, water rushes into these osmolyte-loaded cells $\rightarrow$ CEREBRAL EDEMA, seizures, and herniation. Children are especially vulnerable. Safe limit: lower Na by no more than $\sim 10\text{--}12$ mEq/L per 24 h (~ 0.5 mEq/L/h in chronic cases).

BOARD TRAP

WRONG answer: 'rapid correction of hypernatremia causes osmotic demyelination (central pontine myelinolysis).' That is the complication of correcting HYPOnatremia too fast. The danger of over-rapid HYPERnatremia correction is the OPPOSITE — cerebral EDEMA. Discriminator: hyperNa corrected too fast $\rightarrow$ edema; hypoNa corrected too fast $\rightarrow$ osmotic demyelination.
